MEHDI BELHAJ

Power Electronics & Control Engineer

High-voltage converters · Motor control · Board design · Aerospace

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`270 / 540 VDC · SiC` `DO-160G · DAL A` `PSIM · LTspice` `MATLAB / Simulink`



AVAILABILITY	EXPERIENCE	CONTRACT	MOBILITY	WORK PERMIT	DRIVING LICENCE	ENGLISH
Immediate	3 years · aerospace	Permanent · Fixed-term · Umbrella	All of France	Valid to Nov. 2028	B	C1 · fluent

PROFILE

Power electronics engineer (MSc in EEA) with 3 years of experience in aerospace (Safran, Airbus) and high-voltage R&D (SuperGrid Institute). I design high-voltage converters — **Flyback supplies, a 10 kW SiC inverter, EMC filters** — and model their control and that of electric motors (**Simulink, PSIM**), from specification to testing, up to DAL A.

SKILLS

POWER ELECTRONICS — **Flyback**, Buck, SEPIC, LDO · **Three-phase SiC inverters** · Isolated gate drivers, DESAT · RCD snubbers · Precharge and inrush limiting · Magnetic components

CONTROL ENGINEERING — **MATLAB / Simulink / Simscape · PI / PID controllers** (pole placement, margins) · Nested current, speed and position loops · Stability (Bode, Nyquist) · BLDC and induction machines

EMC & QUALIFICATION — **DO-160G**: conducted emissions (LISN), lightning, ESD · Common- and differential-mode filters · Insulation at altitude (IEC 60664-1) · Dielectric testing

BOARD DESIGN — **OrCAD Capture CIS**, KiCad · **IPC-2221** layout rules · Component selection and derating · Current, voltage and temperature sensing · DFX

SIMULATION & TOOLS — **PSIM, LTspice**, Saber, Scilab / Xcos · Python, C++ · Test benches, oscilloscope · Windchill PLM

PROFESSIONAL EXPERIENCE

SAFRAN

Power Electronics & Hardware Engineer — Safran Ventilation Systems

`Nov. 2023 – May 2025 · 19 months`

Via Centum T&S · Blagnac, France · EMC DO-160G category H

270 VDC mock-up of an aircraft fan originally designed for 115 VAC: 1,890 W input board (−40 to +100 °C) supplying the inverter and the motor control boards.

- **Design**: 9 justified functions, including reverse-current protection, precharge (DO-160G inrush envelope), phase-loss detection in under 4 ms, 200 ms power hold-up, lightning and 15 kV ESD protection, EMC filter.
- **Isolated Flyback supply**: multi-output (132 kHz) from the 270 VDC bus, two isolation groups; 5, 16 and 18 V rails for the control boards via LDOs and a 1.5 MHz Buck.
- **EMC filter**: damped differential- and common-mode stages, sized from a simulation of the 20 kHz PWM inverter and the motor (load steps, FFT).
- **6-layer PCB**: layout specification — IPC-2221 clearances corrected for altitude, DC-bus traces rated 9 A RMS, DFX.
- **Deliverables and planning**: DJD, calculation notes, bench debugging; schedule, milestones, weekly meeting in English with Safran USA.

Tools: PSIM (inverter, motor, FFT) · OrCAD Capture CIS · Excel (calculation notes, EMC filters) · Windchill PLM

Results: 270 VDC feasibility demonstrated with minimal impact on software and mechanics.

AIRBUS

Power Electronics & Control Engineer — Airbus / Crouzet

`Sept. 2022 – Nov. 2023 · 15 months`

Via Serma Ingénierie · Toulouse · R&D, Zero Emission programme · DAL A · ABD0100

Control electronics of an electromechanical spoiler actuator deployed in under 130 ms against gusts: 3 boards (CPU, HVDC, inverter) in 65 × 65 × 170 mm.

- **Inverter**: three-phase 10 kW SiC inverter for a BLDC motor (6 × 1200 V MOSFETs, UCC21750 drivers with DESAT, phase-current sensing); thermal analysis of SiC junctions under pulsed operation.
- **Control**: mechatronic modelling of the actuator, BLDC motor position control, PI tuning over the acceleration and deceleration phases; inverter–motor simulation.
- **±270 V HVDC board**: EMC filtering, ESD protection, current / voltage sensing, quasi-resonant Flyback 540 V to 24 V, Buck 24 V to 12 V.

- **Specification and safety**: contribution to the requirements specification, sub-assembly specifications, justification files; requirements with the Safety team (Serma Safety).

  Tools: MATLAB / Simulink / Simscape · Scilab / Xcos · PSIM · LTspice · OrCAD · KiCad

Results: 130 ms requirement validated in simulation; hardware architecture approved in Airbus and Crouzet design reviews.

Final-year internship, R&D power electronics — SuperGrid Institute `Feb. – Jul. 2022 · 6 months`

Villeurbanne, France · research institute for HVDC grids

Isolated power supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs.

- **Design**: cascade of 12 ferrite transformers (TDK EPCOS T38), trade-off between losses, insulation and size; model validated with oscilloscope measurements.
- **EMC and testing**: common-mode currents under 16 kV/µs edges and countermeasures; dielectric testing.

  Tools: LTspice · oscilloscope · high-voltage dielectric test bench

Results: 20 kV insulation validated, industrialisation specifications delivered.

UNIVERSITY & PERSONAL PROJECTS

- **Induction machine control** (MSc EEA, 2020): simulation of the machine and its control, nested current, flux and speed loops, limiters, robustness tests. **Tools**: Saber
- **Switch-mode power supply** (MSc EEA, 2021): peak current-mode control and voltage loop, filtering, performance measurements.
- **BLDC motor on FPGA** (ENSEEIHT, 2021): commutation and PWM modelled, PWM and speed measurement coded in VHDL on a DE0-Nano board (Cyclone IV), NIOS II processor, control law discretised and coded in C with interrupts; speed control validated on a real motor. **Tools**: PSIM · MATLAB · Quartus · Qsys
- **Python — test automation** (scope-emi-toolkit): parsing of oscilloscope CSV exports (Tektronix, Rigol), windowed spectrum calibrated in dBµV, comparison with a 150 kHz – 152 MHz conducted-emissions mask and pass / fail report; 8 unit tests, continuous integration.

  Tools: NumPy · SciPy · Matplotlib · PyTest · GitHub Actions

ENTREPRENEURSHIP

Co-founding partner — TOKYOSEAT `Since Jul. 2023`

SAS with 5 partners · side activity

- Product compliance for electrical equipment sold in 6 European countries: CE marking (Low Voltage, EMC), RoHS, WEEE, UKCA, laboratory testing and technical files.

EDUCATION & CERTIFICATIONS

Master's degree in EEA — Power Electronics, Actuators & Control · Université Toulouse III Paul Sabatier · INP-ENSEEIHT `2020 – 2022`

Bachelor's degree in EEA · Université Toulouse III Paul Sabatier `2019`

Two-year technical degree (DUT) in Electrical Engineering and Industrial Computing · ENSAM Rabat, Morocco `2018`

C++ and Python certifications — Cisco Networking Academy / OpenEDG `2025 – 2026`

C++ Essentials 1 and 2, C++ Advanced, Python Essentials 1 and 2 · 330 h



French electrical safety authorisation H0, B2V, B2V Essai, BE Essai — CAMIRA Formation `2022`

EYE Berlin — European Youth Event, European Parliament · certificate of participation `Apr. 2024`

LANGUAGES

French bilingual · **English** C1, professional use · **Spanish, Portuguese** basic

SOFT SKILLS & INTERESTS

Autonomous · rigorous · methodical · team player · Competitive football · NQT mentoring programme · Travel (18 countries)